Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by total mortality class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (101 sites, 361 site-years) · deadtrees v5

ΔRMSE = RMSE(with M) − RMSE(without M)

GBoost (Optuna) · 12r | GBoost (Optuna) · 24r | ghtGBM (Optuna) · 12 | ghtGBM (Optuna) · 24 | toGluon (ensemble) · 1 | toGluon (ensemble) · 2

GPPsat (µmol m⁻² s⁻¹): 57% improved | 58% improved | 56% improved | 55% improved | 57% improved | 64% improved

NEPmax (µmol m⁻² s⁻¹): 68% improved | 70% improved | 74% improved | 68% improved | 69% improved | 65% improved

ETmax (mm d⁻¹): 58% improved | 64% improved | 58% improved | 57% improved | 57% improved | 62% improved

WUE (g C mm⁻¹): 53% improved | 58% improved | 53% improved | 51% improved | 55% improved | 48% improved

Sites (sorted by ΔRMSE within each panel)

Total mortality
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 10.6 %, 41 sites)
- Mid (10.6–24.0 %, 38 sites)
- High (> 24.0 %, 22 sites)